

MIA MORRISON



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Website

PROFESSIONAL SUMMARY

Aerospace Engineering student with experience in robotics, mechanical design, systems integration, and technical project development. Recognized statewide leader and two-time Florida Youth of the Year with a proven record of developing programs, leading teams, organizing large-scale initiatives, and communicating complex ideas to diverse audiences. Passionate about applying systems thinking, engineering design, and leadership to solve real-world challenges.

EDUCATION

EMBRY-RIDDLE AERONAUTICAL UNIVERSITY (ERAU) | B.S. Aerospace Engineering | GPA: 3.33

Est. May 2027

ENGINEERING WORK EXPERIENCE

DOCKING ROBOTICS | MECHANICAL ENGINEERING INTERN | SolidWorks

May 2025 – Sept 2025

- Designed and 3D-printed a modular drone landing dock in SolidWorks using a six-panel interlocking system optimized for strength, precision, and rapid assembly.
- Conducted material selection, stress analysis, and torque calculations to improve mechanical performance; currently integrating servo-based rotational actuation for automated movement.

TECHNICAL PROJECTS

PATHFINDER AODR-01 AUTONOMOUS ROVER | Arduino, CAD, 3D Printing, PCBWay Sponsorship

June 2026 – Present

- Independently designed and built an autonomous obstacle-detection rover using an Arduino Nano, ultrasonic sensor, DC motors, LCD display, and custom 3D-printed chassis, with self-taught Arduino programming and embedded systems.
- Selected for a PCBWay manufacturing sponsorship; redesigned prototype currently in professional production.**

PYTHON Satellite Trajectory Analysis Tool | Python, Tkinter, NumPy, Matplotlib, PyAstronomy, SGP4

July 2026

- Independently developed an interactive desktop application for satellite trajectory analysis and orbit visualization using Python.
- Implemented real-time 3D orbit visualization through adjustable Keplerian orbital elements and generated altitude profiles from TLE data using the SGP4 propagator. Integrated data visualization, GUI development, and CSV export.

MATLAB ACTIVE RECALL STUDY SYSTEM

April 2026

- Developed a MATLAB study guide program that presents questions, evaluates responses, provides immediate feedback, and explains solutions to reinforce understanding of course material.

MATLAB MEAL PREP SYSTEM

February 2024 – April 2024

- Developed a MATLAB-based application that recommends high-protein, low-calorie recipes based on user meal preferences, protein selection, and available cooking time. Utilized an Excel-based recipe database and filtering logic.

CATIA SCOOTER DESIGN

Sept 2023 – Dec 2023

- Reverse-engineered and modeled a complete electric scooter assembly using CATIA.
 - Created detailed engineering drawings, technical documentation, and assembly models.
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Software Skills: SolidWorks, Onshape, CATIA, MATLAB, Arduino IDE, Python, Microsoft 365

Leadership Skills: Public Speaking, Project Management, Team Leadership, Youth Mentorship, Community Engagement

LEADERSHIP & COMMUNITY EXPERIENCE

FAITH IN FLORIDA- INDIVIDUAL DONOR COORDINATOR | CIVIC ENGAGEMENT ORGANIZER

April 2021- Present

- Lead donor stewardship initiatives, fundraising campaigns, and community engagement efforts for statewide civic programs.
- Organized candidate forums, voter registration drives, teach Black history and college readiness to youth throughout Florida.

NATIONAL PUBLIC SPEAKER, YOUTH ADVOCATE (BGCA)

June 2022-Present

- Advocated for mental health awareness and educational resources nationally as the 2022 & 2023 Florida Youth of the Year.